

Assignment 2

Turning FinGPT into a

Quantitative Signal Module

Transform your fine-tuned FinGPT from a general financial LLM

into a structured signal generator for quantitative pipelines.

ASSIGNMENT	TRACK	DUE DATE	DELIVERABLES
02 / 02	FinGPT LLM	April 20, 2026	Blog + GitHub

GITHUB REPOSITORY

github.com/AI4Finance-Foundation/FinGPT

01 Objectives

GOAL ONE

Improve FinGPT as a Financial Analyst

Adapt your fine-tuned FinGPT model to behave as a stronger financial analysis engine — especially for sentiment understanding, short-term price direction prediction, and financial news interpretation.

GOAL TWO

Convert Output into Quantitative Signals

Design a method that transforms FinGPT free-text outputs into numerical indicators or structured signals that can be consumed by a quantitative pipeline for ranking, filtering, forecasting, or decision support.

02 Assignment Tracks

TRACK A FinGPT as a Better Analyst

Improve the model's financial understanding and prediction quality. The goal is to make FinGPT a more reliable and consistent financial analysis engine — not just a chatbot that generates fluent text.

Possible directions:

Better prompting · Output formatting · Fine-tuning design · Post-processing · Calibration

TRACK B FinGPT as a Signal Generator

Convert model outputs into structured forms that a downstream quant strategy can consume. The signal must be consistent, measurable, and convertible for use in a real trading or research pipeline.

Possible directions:

Sentiment score · Direction label · Confidence score · Event-impact score · Composite signal

Key principle: The goal is not to build a full trading system. The goal is to make FinGPT behave as a modular signal component — producing outputs that are consistent, measurable, and useful for downstream quantitative decisions.

03 Module Design Overview

In quantitative finance, a language model must behave less like a chatbot and more like a modular component. This means its outputs must be consistent, measurable, and convertible into signals. FinGPT is positioned as a plug-in analysis module for quantitative use — the key challenge in this assignment is to make that positioning real, by designing structured output schemas, text-to-signal mappings, and calibration methods that make FinGPT genuinely useful inside a systematic investment workflow.

3.1 The Four Specialized Modules

A

PRICE TREND PREDICTOR

Use the model directional judgment as a trading signal. Improve accuracy, consistency, and practical usability.

B

SENTIMENT SCORER

Convert qualitative output into a structured sentiment score (-2 to +2). Make it robust, repeatable, and interpretable.

C

NEWS ANALYST

Process news and generate: event type, sentiment label, urgency score, and expected market impact.

D

YOUR OWN IDEA

Go beyond the listed options with a well-motivated idea for extracting quant signals from a financial LLM.

3.2 Text-to-Signal Mapping

SIGNAL TYPE

Sentiment Score

RANGE

-2 → 0 → +2

DIRECTION

Bullish / Neutral / Bearish

CONFIDENCE

Probability-like score

EVENT IMPACT

Severity over time window

COMPOSITE

Multi-factor combined score

3.3 Suggested Extensions

5.1

Confidence Calibration

Calibrate output confidence using self-consistency, repeated prompting, or score normalization. Unstable confidence degrades signal quality.

5.2

Structured Output Design

Force the model to generate a fixed schema: sentiment label, score, confidence, risk factors, forecast horizon, and signal explanation.

5.3

Multi-Signal Design

Produce richer outputs: sentiment score, event surprise, confidence, volatility warning, and short-term direction in a single pass.

5.4

Error Analysis

Analyze when the model fails — high-volatility periods, headline overreaction, or confusing long-term fundamentals with short-term signals.

04 What You Should Do

1

Adapt Your Fine-Tuned FinGPT Model

Take your fine-tuned model from Assignment 1 and extend it into a module that serves a quantitative research or trading pipeline. Treat it as a financial analysis engine with machine-readable outputs, not a pure text generator.

2

Improve Prediction Accuracy

Improve the model's ability to predict future price trend direction. Explore cleaner output templates, confidence-aware outputs, constrained label spaces, better prompt instructions, and ensemble-style post-processing.

3

Design a Text-to-Signal Mapping Method

Design a robust method to convert model-generated text into a numerical or categorical signal for downstream use — sentiment score, direction label, probability-like confidence score, or multi-factor composite signal.

4

Explore at Least One Specialized Module

Seriously implement at least one of the four directions: Price Trend Predictor, Sentiment Scorer, News Analyst, or your own well-motivated idea. You are encouraged to explore more than one.

5

Extensions and Enhancements [optional]

Consider confidence calibration, structured output schemas, multi-signal design, or error analysis to make your module more rigorous and research-grade.

05 Deliverables

A	B	C
MEDIUM BLOG POST A Medium-style blog post on how you structured FinGPT as a reusable module, which methods you chose, what you implemented, and what you learned. module design · methods · results	GITHUB CODE Upload code to GitHub, organized so someone can understand what the module does, how it produces signals, and how outputs are reused in a quant pipeline. clean repo · signal logic · reusable	OPTIONAL EXTRAS Additional experiments, calibration results, error analysis, or multi-signal designs beyond the minimum requirement are strongly encouraged. calibration · error analysis · extensions

5.1 Recommended Report Structure

1	Introduction Purpose of the assignment and why turning an LLM into a signal module matters.	5	Results Performance improvements or signal quality gains you observed.
2	Module Design How you structured FinGPT as a reusable component inside a quant pipeline.	6	Analysis What worked, what did not, where the model still fails, which design choices mattered.
3	Methodology Which methods you implemented, output format design, and text-to-signal mapping.	7	Optional Extra ideas, experiments, or improvements beyond the minimum requirement.
4	Experiments Trend prediction, sentiment scoring, news analysis, or output format comparisons.	8	Conclusion Main lessons from the assignment.

06 Evaluation Criteria

#	Criterion	Description	Weight
01	Module Design Quality	Did you successfully turn FinGPT into a reusable analysis component rather than a generic chatbot?	
02	Signal Design	Did you design a sensible and usable method for converting language outputs into structured signals?	
03	Financial Relevance	Are your outputs meaningful for quantitative finance or trading-related use?	
04	Technical Depth	Did you thoughtfully improve prompting, fine-tuning, post-processing, or output structure?	
05	Analytical Rigor	Did you clearly explain your methodology, experiments, and findings?	
06	Originality	Did you go beyond the minimum and explore new ideas where appropriate?	